

Table 4. Effect of Prompt Variants on Generated Test Input Diversity.

| Agent #  | Prompt 1<br>(No constraints) |            |                       | Prompt 2<br>(Tool Constraints) |            |                       | Prompt 3<br>(Ambiguity Constraints) |            |                       | Prompt 4 (AgentInspect)<br>(Tool + Ambiguity Constraints) |            |                       |
|----------|------------------------------|------------|-----------------------|--------------------------------|------------|-----------------------|-------------------------------------|------------|-----------------------|-----------------------------------------------------------|------------|-----------------------|
|          | 1-step                       | Multi-step | Tool Pattern Coverage | 1-step                         | Multi-step | Tool Pattern Coverage | 1-step                              | Multi-step | Tool Pattern Coverage | 1-step                                                    | Multi-step | Tool Pattern Coverage |
| Agent 1  | 15                           | 12         | A, S, M               | 1                              | 27         | P, S, M               | 12                                  | 17         | A, S, M               | 1                                                         | 25         | P, S, M               |
| Agent 2  | 17                           | 13         | P, S                  | 12                             | 18         | A, S, M               | 23                                  | 7          | A, S                  | 2                                                         | 28         | A, S                  |
| Agent 6  | 21                           | 4          | A, S                  | 21                             | 4          | A, S                  | 22                                  | 4          | A, S                  | 11                                                        | 4          | A, S                  |
| Agent 7  | 2                            | 6          | A, S                  | 5                              | 16         | A, S, M               | 9                                   | 17         | A, S, M               | 2                                                         | 25         | A, S, M               |
| Agent 13 | 1                            | 29         | A, S                  | 0                              | 30         | A, S                  | 0                                   | 30         | A, S                  | 0                                                         | 30         | A, S                  |
| Agent 16 | 16                           | 14         | A, S                  | 23                             | 7          | A, S                  | 26                                  | 4          | A, S                  | 16                                                        | 3          | A, S                  |
| Agent 17 | 24                           | 6          | A, S                  | 9                              | 21         | A, S                  | 21                                  | 9          | A, S                  | 19                                                        | 11         | A, S                  |
| Agent 20 | 14                           | 16         | A, S                  | 10                             | 20         | A, S                  | 17                                  | 13         | A, S                  | 10                                                        | 20         | A, S                  |
| Agent 25 | 15                           | 0          | A                     | 6                              | 3          | A, S                  | 11                                  | 6          | A, S, M               | 0                                                         | 10         | A, S, M               |
| Agent 28 | 30                           | 0          | A                     | 30                             | 0          | A                     | 30                                  | 0          | A                     | 30                                                        | 0          | A                     |
| Agent 29 | 12                           | 10         | A, S, M               | 6                              | 15         | A, S, M               | 5                                   | 9          | A, S, M               | 5                                                         | 16         | A, S, M               |
| Agent 31 | 5                            | 10         | A, S, M               | 0                              | 13         | P, S, M               | 7                                   | 13         | A, S, M               | 2                                                         | 21         | A, S, M               |

A: all integrated tools, P: partial integrated tools, S: same-tool chaining, M: multi-tool chaining